

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AAE5209
Subject Title	Computational Fluid Dynamics
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	1. To familiarise students with computational fluid dynamics (CFD) techniques used in the study of fluid flow problems. 2. To enable students to apply CFD techniques specifically to aerospace-related scenarios.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. acquire fundamental knowledge of CFD techniques; b. perform numerical analysis of canonical flow problems; and c. gain basic understanding of state-of-the-art CFD techniques.
Subject Synopsis/ Indicative Syllabus	Revisit to flow governing equations – Material derivative; Reynolds transport theorem; Derivation of Navier-Stokes equations. Turbulent flows – Kolmogorov scales; Reynolds-averaged Navier-Stokes simulations; Large-eddy simulation. Introduction to numerical methods – Errors; Consistency; Accuracy; Stability; Convergence. Finite-difference methods – Discretisation of temporal and spatial derivatives; Explicit and implicit formulations; Stability analysis; Fourier error analysis. Time-marching techniques – Lax–Wendroff technique; MacCormack’s technique; Crank–Nicolson technique; Range–Kutta method. Finite-volume methods – Principles of finite volume method (FVM); FVM for diffusion and convection problems; Application to Navier-Stokes equations, SIMPLE algorithm, PISO algorithm. Modern CFD techniques – Shock-capturing techniques; Upwind schemes; Limiters; Total variation diminishing; Implicit methods. Case studies – Application of the numerical techniques to canonical flow problems in aerospace engineering.

Teaching/Learning Methodology	1. The teaching and learning methods include lectures/tutorials, projects, and homework assignments. 2. The lectures/tutorials aim at providing students with integrated knowledge of CFD. 3. Technical/scientific problems and examples will be raised in projects and homework assignments to develop students’ skills of numerical analysis. <table><tr><td rowspan="2">Teaching/Learning Methodology</td><td colspan="3">Intended subject learning outcomes</td></tr><tr><td>a</td><td>b</td><td>c</td></tr><tr><td>1. Lectures/tutorials</td><td>√</td><td>√</td><td>√</td></tr><tr><td>2. Projects</td><td>√</td><td>√</td><td>√</td></tr><tr><td>3. Homework assignments</td><td>√</td><td>√</td><td>√</td></tr></table>	Teaching/Learning Methodology	Intended subject learning outcomes			a	b	c	1. Lectures/tutorials	√	√	√	2. Projects	√	√	√	3. Homework assignments	√	√	√														
Teaching/Learning Methodology	Intended subject learning outcomes																																	
	a	b	c																															
1. Lectures/tutorials	√	√	√																															
2. Projects	√	√	√																															
3. Homework assignments	√	√	√																															
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><td rowspan="2">Specific assessment methods/tasks</td><td rowspan="2">% weighting</td><td colspan="3">Intended subject learning outcomes to be assessed (Please tick as appropriate)</td></tr><tr><td>a</td><td>b</td><td>c</td></tr><tr><td>1. Attendance</td><td>5%</td><td>√</td><td>√</td><td>√</td></tr><tr><td>2. Assignment</td><td>20%</td><td>√</td><td>√</td><td>√</td></tr><tr><td>3. Projects</td><td>20%</td><td>√</td><td>√</td><td>√</td></tr><tr><td>4. Final exam</td><td>55%</td><td>√</td><td>√</td><td></td></tr><tr><td>Total</td><td>100%</td><td colspan="3"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: 1. The assessment is comprised of 45% continuous assessment (projects and tests) and 55% examination. 2. The continuous assessment consists of assignment and projects. They are used to evaluate the progress of students’ study, assist them in self-monitoring of fulfilling the respective subject learning outcomes, and enhance the integration of the knowledge learnt. 3. The examination is used to assess the knowledge acquired by the students for understanding and analysing the problems critically and independently; as well as to determine the degree of achieving the subject learning outcomes.	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			a	b	c	1. Attendance	5%	√	√	√	2. Assignment	20%	√	√	√	3. Projects	20%	√	√	√	4. Final exam	55%	√	√		Total	100%			
Specific assessment methods/tasks	% weighting			Intended subject learning outcomes to be assessed (Please tick as appropriate)																														
		a	b	c																														
1. Attendance	5%	√	√	√																														
2. Assignment	20%	√	√	√																														
3. Projects	20%	√	√	√																														
4. Final exam	55%	√	√																															
Total	100%																																	
Student Study Effort Expected	<table><tr><td>Class contact:</td><td></td></tr><tr><td>▪ Lecture</td><td>33 Hrs.</td></tr><tr><td>▪ Tutorial</td><td>6 Hrs.</td></tr></table>	Class contact:		▪ Lecture	33 Hrs.	▪ Tutorial	6 Hrs.																											
Class contact:																																		
▪ Lecture	33 Hrs.																																	
▪ Tutorial	6 Hrs.																																	

	Other student study effort:	
	▪ Self-learning	33 Hrs.
	▪ Homework	50 Hrs.
	Total student study effort	122 Hrs.
Reading List and References	1. Anderson D. A., Tannehill, J. C., Pletcher R. H., Munipalli R., and Shankar V., Computational Fluid Mechanics and Heat Transfer. CRC Press, 4th edition, 2020. 2. Anderson J. D., Computational Fluid Dynamics: The Basics with Applications. McGraw-Hill, 1st edition, 1995. 3. Ferziger J. H., Perić M., and Street R. L., Computational Methods for Fluid Dynamics. Springer, 4th edition, 2019.	

May 2025